

Experiment -1:- Relational DataBase Management Systems (RDBMS)

Various versions of Oracle Database

- * Oracle 6 : 6.0.17 - 6.2.0
- * Oracle 7 : 7.0.12 - 7.3.4.5
- * Oracle 8 : 8.0.3 - 8.0.6
- * Oracle 8i : 8.1.5.0 - 8.17.4
- * Oracle 9i : 9.0.10 - 9.0.1.4
- * Oracle 9i : 9.2.0.1 - 9.2.0.8
- * Oracle 10g : 10.1.0.2 - 10.1.0.5
- * Oracle 10g : 10.2.0.1 - 10.2.0.5
- * Oracle 11g : 11.1.0.6 - 11.1.0.7
- * Oracle 11g : 11.2.0.1 - 11.2.0.4
- * Oracle 12c : 12.1.0.1 - 12.1.0.2
- * Oracle 12c : 12.2.0.1 - 12.1.0.2
- * Oracle 18c : 18.1.0 - 18.9.0
- * Oracle 19c : 19.1.0 - 19.6.0
- * Oracle 21c : 21.1.0
- * Oracle 23c : 23.2.0

History of DBMS

- * In 1960 → Integrated Data Structure (IDS)
- * Late 1960's → Information Management System (IMS)
- * 1970 → Relational model followed by RDBMS.
- * 1974 → IBM developed prototype System: R

* 1976 → Multics → first released RDBMS by bell labs & Honey bell.

* 1979-80 → Oracle RDBMS → Relational Software Company.

* 1984 → IBM developed language called SEQUEL renamed as SQL designed for only RDBMS.

Popular RDBMS Include :-

1. Oracle
2. MySQL
3. Microsoft SQL Server
4. PostgreSQL
5. SQLite
6. Maria DB
7. IBM Informix
8. Apache Derby
9. H2 database Engine
10. Amazon Relational Database Server
11. Sybase ASE
12. DB2
13. Microsoft Access.

* SQL: (Structured Query Language) - A language to access data in RDBMS.

* Language like SQL :-

1. X Query
2. OQL → Object Query Language
3. GraphQL
4. LINQ → Language Integrated Query.

SQL:- SQL language supports a number of commands known as database language command.

Query:- A Query is a statement return in SQL to work with database.

DataBase Language Commands :-

① Data Definition Language Command (DDL):- The DDL Commands are used for create, modify and delete various data objects like tables, views, triggers and constraints.

→ These are used by DBA.

⇒ The Commands under these category are :-

1. Create Command : Used to create tables (or) views.
 2. alter Command : Used to change the schema of tables (or) views.
 3. drop Command : Used to delete data objects from database.
 4. Truncate Command : Used to delete all the records and this schema from the databases.
 5. Rename Command : Used to rename the existing table.
- ② Data Manipulation Language Command (DML) :- Used for manipulate on of data in database.
1. Insert Command : Used for inserting rows of data
 2. update Command : Used to modify existing data.
 3. delete Command : Used to delete existing data.
 4. merge Command : Used to insert (or) update data from multiple tables
 5. call Command : Used to call PLSQL procedures (or) Java functions from database.
 6. Lock table Command : Used to Lock a table for concurrency control.
- ③ Data Control Language Command (DCL) :- These Commands are used by DBA to control access of data by various users in the database.
1. Grant Command : Used to provide access permission for various database user.
 2. Revoke Command : Used to remove the access privileges of the database users.
- ④ Data Query Language Command (DQL) :- These Commands are used for retrieving the data from the database under this we have only single Command name 'select' Command.
- ⑤ Transaction Control Language Command (TCL) :- These Commands are used for controlling the operations in the Transaction.

1. Commit Command: This is used to save the workdone over the database.
2. Rollback Command: It restores the database to the original content since the last commit command.
3. Savepoint Command: Used to define a point in a transaction to which we can perform roll backing.

* SQL is not a case-sensitive language we can use both Capital and small letters.

Data-Types in Oracle :-

1. Char Datatype - This datatype is used to store characters (or) strings of fixed length.

Syntax:- char(size)

ex:- name char(10);

2. Varchar (or) Varchar2 - This datatype is used to store variable length alphanumeric data

Syntax:- Varchar(size) (or) Varchar2(size)

3. Number - Used to store integers and floating point numbers.

Syntax:- number(p), number(p,s) P → Precision, S → Scale

ex:- number(8,2)

* It is going to store the 8 bit of integral part and 2 bits for storing the floating number.

4. Date - Used to store (or) represent the date and time for the content stored in the database.

Declaration: Variable name date;

Default Syntax: dd-mm-yy

Experiment-2 :- Practicing SQL Commands with Syntaxes and examples.

(DDL - create, alter, rename, truncate, drop; DML - insert, update, delete;
DCL - grant, revoke; TCL - rollback, commit, savepoint; DQL - select)

Program:-

DDL → ①. Create Command:- To create a table 'create' Command is used for creating tables in Oracle.

Syntax:-
Create table tablename (Col1 datatype(size), Col2 datatype(size),
Coln datatype(size));

→ For the above example: creating a employee table:

Create table employee (ename char(50), eid varchar(10), location char(10),
phoneno number(10));

Output:- table created

Student table Schema:- student(name, rollno, branch, marks, dob)

Creation:- create table student (name varchar(10), rollno varchar(10),
branch char(10), marks number(10), dob Date);

Output:- table created

2. Alter Command:- The schema of a table can be modified by using 'alter' Command.

→ Alter Command can be used to add set of columns (or) delete set of columns, ~~for delete~~ set modify size of attributes and can change the names of existing columns.

Syntax: 1 Adding the attributes to the Schema

alter table <table-name> add (new column name1 datatype(size), new column name2 datatype(size));

Example-1:- Change the schema of student table by adding location

alter table student add(location varchar(10));

output:- table altered.

→ desc student;

output:-

name	type
name	varchar(10)
rollno	varchar(10)
branch	char(10)
marks	number(10)
dob	date
location	varchar(10)

Example-2:- Alter the employee table by adding hra and da attributes.

Syntax:- alter table employee add(da number(10), hra number(10));

output:- table altered.

desc employee;

output:-

name	type
EID	varchar(10)
ename	varchar(10)
Designation	char(10)
Salary	number(10)
Dob	date
location	char(10)
DA	number(10)
HRA	number(10)

3. Drop Command (Deleting a Column) :-

Syntax:- `alter table <tablename> drop column Column_name;`

Example-1:- Alter the employee table by deleting hra and da attributes.

`alter table employee drop column (hra, da);`

Output:- table altered

`desc employee;`

Output:-

name	type
EMPID	Varchar(10)
ename	Varchar(10)
Designation	char(10)
Salary	number(10)
DOJ	Date
Location	Char(10)

Example-2:- Alter the student table by modifying (or) deleting location attribute.

`alter table student drop column location;`

Output:- table altered

`desc student;`

Output:-

name	type
Roll no.	Varchar(10)
name	Varchar2(10)
Branch	char(10)
Marks	number(10)
DOB	Date

Modifying the size of existing datatypes:-

Syntax:- alter table tablename modify columnname newtype(size);

Ex:- modify the size of eid in employee table

Syntax:- alter table employee ^{modify} eid varchar(15);

Output:- table altered

desc employee;

Output:-

name	type
EID	varchar(15)
Ename	varchar(10)
Designation	char(15)
Salary	number(10)
DOJ	Date
Location	char(10)

Renaming table:- we can use rename command to rename the existing table.

Syntax:- rename oldtable_name to newtable_name;

Example:- rename employee to emp;
rename student to StudentsInfo;

Drop and Truncate Commands:- These are used for deleting the contents from a table.

- When we use the truncate command, it deletes the set of rows present in a table, the structure (or) schema of the table will be existing.
- When we use drop command it deletes all the rows present in the table and

the schema, includes triggers.

- Once a table is dropped, we cannot get it back.

4. Drop:-

Syntax:- drop table tablename;

Ex:- Create table library (book_no number(10), no_of_books number(10));

Output:- table created

⇒ drop table library;

Output:- Table dropped.

5. Truncate:-

Syntax:- truncate table tablename;

Ex:- truncate table dept;

Output:- table truncated

→ desc dept;

Output:-

name	type
Dept ID	varchar(10)
DeptName	varchar(10)
Location	char(10)

DML Commands → 1. Insert Command:-

Syntax:- insert into tablename values (values separated with commas);

Inserting into employee table:-

Insert into employee values ('ramya', '22e1', '22-apr-17', '67889', 'hyd', 'Srmmanager');

Output:- 1 row inserted.

Inserting into student table:-

insert into student values('Tina', '2291', 'cse', '78', '22-apr-05');

Output:- 1 row inserted.

Syntax 2:- insert into tablename(column_names) values(separated by commas);

Example:- inserting into customer table:-

insert into customer(cname, ~~loc~~ location, phone_no) values('abc', 'hyd', 222);

Output:- 1 row created

Syntax 3:- Interactive insertion

insert into <table name> values(&col1, &col2, &col3, &col4, &coln);
enter the value of &col1: ' ' 1

⋮
enter the value of &coln: ' ' 1 ✓

Example:- insert the data into employee table

insert into employee values(&ename, &eid, &dof, &salary, &location, &designation);

Output:- enter the ename: 'ramya'

enter the eid: '22e1'

enter the dof: '22-apr-17'

enter the value for salary: 67889

enter the value for location: 'hyd'

enter the value for designation: 'Snr Manager'.

2. update Command :- update command is used to change the data values present in a table.

- The update can be used for updating all the rows (or) set of rows.

Syntax :- Updating all the rows (without condition)

update tablename set column_names = value, column_name2 = value,;

Query :- *Update the salaries of employees by giving 1000 rupees as bonus.

update employee set salary = salary + 1000;

Output :- table updated

*Update the marks of students by giving 10 extra marks of students

update students set marks = marks + 10;

Output :- table updated

Syntax-2 :- With Condition

update <table name> set columnname1 = value/expression, columnname2 = value/expression where <condition>;

Example :- For employee table

Update the salaries of employees by giving ~~height~~ ^{hike} of 10% for developers.

Query :- update employee set salary = salary + (salary * 10) / 100 where designation = 'Snr_manager';

Output :- 1 row updated.

3. Delete Command :- The delete command can be used for deleting set of rows from a table.

Syntax :- delete from tablename;

The above command is used for deleting all the set of rows present in the table.

Ex:- delete from library;

Output:- 2 rows deleted;

Syntax2:- Deleting based on a condition

Syntax:- delete from tablename where condition;

Example:- delete from employee where Salary $\geq$ 50000 and location = 'hyd';

Output:- 3 rows deleted;

DQL Commands:- Retrieving the data from tables (select)

Select Command can be used for retrieving data from one (or) more tables.

Syntax1:- Selecting all rows of table

Syntax:- select * from tablename;

Query:- Display the details of the employee:

select * from employee;

Output:-

Ename	Eid	DoJ	Salary	location	Designation
Ramya	22e1	22-apr-15	67889	Srd	SrT manager
MusVira	22e5	22-apr-15	676392	hyd	director
harish	22e7	07-Mar-23	67431	hyd	program manager
abhi	22657	11-Jan-24	76431	khammam	Administrative Officer
bhavana	2265e4	20-Jan-23	54312	hyd	Jnr manager

Syntax 2:- To display the content of few columns of a table.

Syntax:- select column1, col2, from table name;

Display the employee details such as name, eid, salary.

Query:- select eid, ename, salary from employee;

Output:-

ename	eid	salary
Ramya	22e1	67889
Musvira	22e5	67839
harish	22e7	67431
abhi	22657	76431
bhavana	2265e4	54312

Syntax 3:- select * (or) columnname from tablename((where condition) (group by) attribute 3 (order by attribute));

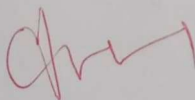
Display the details of a students who got more than 97 marks;
select * from student where marks > 80;

Output:-

name	rollno	Branch	marks	Dob
Premdas	22g7	eee	90	09-mar-04
Seema	22h7	cse	89	09-Jun-04

Commit:- Commit;

Output:- Commit Complete;



Experiment-3:- Operators in SQL

1. Arithmetic Operators:- $+$, $-$, $*$, $/$, $**$, $\%$
2. Relational Operators:- Used for Comparison of 2 expressions when the output of the comparison will be True (or) False.

$<$, $>$, $>=$, $=$, $!=$

Example:- ($=$)

1. Question:- Find the names of the emp whose designation is 'Manager'

Query:- Select name from emp where Designation = 'Manager';

Output:-
ramya
Musvira
Bhavana

2. ($!=$)

Question:- Find the names of the emp who are not living in hyd.

Query:- Select name from emp where location $!=$ 'hyd';

Output:-

Ename
ramya
abhi
harish
bhavana

3. ($>=$, $<=$)

Question:- Find the details of the emp whose salary more than 20k and below 500k.

Query:- Select * from emp where salary $\geq$ 20000 and salary $\leq$ 500000;

Output:-

Ename	Empid	Salary	Location
ramya	2201	74678	Srd
Bhavana	7482	75187	Suryapet

3. Logical Operators :- Used to combine two (or) more conditions And, Or, not.

Examples :- 1. Display the names and id's of the employees whose salary is more than 30k and resides at hyd.

Query :- select ename, eid from emp where Salary > 30000 and location = 'hyd';

Output :-

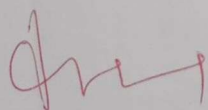
ename	eid
muskan	22e2
musvira	243e7

2. Display the names and id's of the employee whose salary is more than 50k (or) resides in hyd.

Query :- select ename, eid from emp where Salary > 50000 (or) location = 'hyd';

Output :-

ename	eid
muskan	22e2
musvira	22e5
harish	2265e
Ranya	22e1



Experiment-4:- Special Operators in SQL

The various special operators are :-

- | | | |
|----------------|-------------|----------------|
| 1. in | 5. is null | 9. All |
| 2. not in | 6. not null | 10. exists |
| 3. between | 7. like | 11. not exists |
| 4. not between | 8. any | 12. not like |

1. Between:- The operator will return set of rows whose value is in between an inclusive of the specified range.

Example:- Display the details of employee whose salary is in the range of 25k to 38k 750k

Query:-

Select * from ~~range~~ emp where salary between 25000 and ~~75000~~ ⁷⁵⁰⁰⁰⁰

Output:-

Ename	Eid	Salary	location	Depid
harish	22f8	698770	khammam	3

2. not between:- Will display set of rows whose values are not between the specified range.

Example:- Display the names of the students whose marks are not between 10 to 35.

Query:- Select names from Student where marks ~~are~~ not between 10 and 35;

Output:-

Name
muskan
tina
ravi
musvira
meenu

3. in Operator :- Used to verify the membership of an attribute value in set of values.

Example:- List the names of employee whose designation is manager, director

Query:- select enames from emp ^{where} designation in ('manager', 'director');

Output:-

Ename
Ramya
Musvira
Bharana

4. not-in :- It return all the ~~types~~ tuples which does not match with the values specified in the list.

Ex:- Display the names of the student whose branch not in eee, eee.

Query:- select name from student where branch not in ('eee', 'eee');

Output:-

Name
Muskan
tina
meenii

5. IS null :- It return set of rows for which the value of the specified column is not available.

Ex:- Display the names of students whose dob is null.

Query:- select name from student where dob is null;

Output:-

Name
Musvira
tina

6. Is not null :- It returns set of records (or) rows for which the value of the specified column is not a null value.

Example:- Display the name of the students whose marks are not null.

Query:- Select name from student where marks is not null;

Output:-

name
tina
Premdas
Seema

7. Like Operator:- The Like operator is used for Pattern matching (or) character matching. It supports 2 special characters.

1. underscore (-) : One unknown character

2. (% or d) : It indicates zero (or) more characters.

Example:- 1. List the names of the student has any letter starting as 4 letters and ending with letter 'i'.

Query:- Select ename from emp where ename like '____i';

Output:-

ename
abhi

2. Display the names of the emp who have 2nd character in their name as a (or) S.

Query:- Select ename from emp where ename like '____a%' like '____S%';

Output:-

ename
ramya
Sai

Experiment-5:- Working with Order by Clause

The set of rows which is retrieved from the table can be display either in ascending (or) descending order based on attribute values by using orderby clause.

Syntax:- Select * from tablename [Where class] order by column1, (column2), (desc);

- * By default the sorting order is ascending order.
 - * To display the content in descending order we can use 'desc' keyword
 - * The sorting of data can be done using more than 1 column also.
- Instead of using column names we can use column numbers in order by.

Example:- 1. Display the details of emp in the ascending order of salaries.

Query:- Select * from emp order by salary;

Output:-

Ename	Eid	Salary	Location
Ramya	22e1	74678	2
bharana	4492	75167	3
harish	22f8	698770	3
Musvira	223e7	5089087	1
abhi	22e4	7322198	3
muskan	22e2	7939824	1

2. Display name, rollno and branch of student in the descending order of branch.

Query:- Select name, rollno, branch from student order by branch desc;

Output:-

name	rollno	branch
meenu	45978	met
ravi	nrs56	eee
musvira	459h	eee
muskan	e45	cse
tina	ee45	cse

3. Display eid and did and salary in descending using column number(using *)

Query:-

Select * from emp order by 2,5,3;

output:-

ename	eid	Salary	location	Depid
Ramya	22e1	74678	Srd	2
Muskan	22e2	7439832	hyd	1
abhi	22e4	7322198	zhb	3
harish	2278	640770	khammam	3
musvira	223e7	5089097	hyd	1
bhavana	21482	75187	Sur	3

Experiment - 6 :- Aggregate Functions

The Aggregate function will be applied on single of a table and they will return single value.

1. Max(x) :- This function will gives the maximum value present in the column named (x)

2. Min(x) :- This function will gives the minimum value present in the column named (x).

3. Sum(x) :- Gives the total of values present in the column.

4. avg(x) :- Gives the average of the values present in the column.

5. Count(x) :- Gives the no. of rows present in the table. This function will not count the row with null value.

6. Count(*) :- It display the no. of rows present in the table including null and duplicate values.

7. Count(distinct x) :- It display the number of rows with distinct values for x.

8. Stdder(x) :- Gives the standard deviation for the set of values x consists.

9. Variance(x) :- Gives the variance for the set of value x consists.

Queries using Aggregates :-

1. Display the maximum and minimum salaries of the emp.

Query :- select max(salary), min(salary) from emp;

Output :-

MAX(salary)	MIN(salary)
80000.7	40000.7

2. Find the minimum and maximum of the student marks.

Query:- select max(marks), min(marks) from student;

Output:-

Max(marks)	Min(marks)
90.7	97

3. Display the total salaries and avg salaries of the employee.

Query:- select count(salary), avg(salary) from emp;

Output:-

Count(salary)	avg(salary)
7	58286.0143

4. Find the number of emp who are having salary more than 250000

Query:- select count(salary) from emp where salary > 250000;

Output:-

Count(salary)
7

5. Find the number of emp who are working in the location hyd.

Query:- select count(location) from emp where location = 'hyd';

Output:-

Count(location)
3

6. Find the number of emp who are not managers and belong to dept 2.

Query:- select count(*) from emp where designation != 'manager' and dept = 'dept2';

Output:-

Count(*)
2

7. Find the number of students who are having marks greater than 90 and belongs to CSE.

Query :- select count(*) from student where marks > 90 and branch = 'cse';

Output :-

Count(*)
1

8. Find the total amount of salary paid to 'HR'.

Query :- select sum(salary) from emp where designation = 'hr';

Output :-

Sum(salary)
130000.7